

SEE4996/4997 Final-Year Project

Writing Your Report

1. Preparatory work

- The title for the report should
 - be relatively short
 - describe the main theme of your work
- Draft an outline of the sections of the report and decide on the material that should be in the report

2. Some points to note

- Never assume the reader know what you are thinking
- Always put yourself in the reader's shoes – you produce the report not for yourself but for others to read
- Ask yourself questions such as
 - why did I choose this experimental setup/equipment
 - why did I make these assumptions
 - what may be some of the problems of the instrument(s) and how have I addressed these problems
 - were the instruments calibrated properly
 - why is my equipment design/theory better than others
 - are the results consistent with those from previous studies; if not, why are they different
- The first time you use an acronym, always write out the full words first.
- Section headings should be relatively short and descriptive
- Avoid long sections as well as too many sub-sections
- Use consistent grammatical styles throughout the report (active vs. passive voice, present vs. past tense, etc.)
- Make sure that if you use someone else's text, figure, table or plate/image, or any information from the internet, you cite the source. ***Otherwise, this is considered as plagiarism.***

3. Suggested structure

- *Introduction and objectives*
 - Give a background about the topic of the report.
 - Review previous and important/significant research results related to the topic.
 - Based on these previous results, explain why you want to do the present research and what you hope to accomplish, i.e., the objectives of your research.
 - Give a “road map”, i.e., a description of the different sections in the report.

- *Literature review*
Although you have written a literature review in your Interim Report, you should also include such a review in your report, which can contain additional articles that you have read after you have submitted your Interim Report. The literature review can also serve as a “justification” of why you are doing this piece of research. In addition, the extent of the literature review illustrates your efforts and interest in the subject.
- *Experimental setup/Theoretical framework*
 - For experimental work,
 - ◇ describe the experimental setup/equipment you used in the research
 - ◇ provide at least a simplified theory of the operation of the equipment
 - ◇ explain why you chose such a setup/equipment
 - ◇ if you are designing your own setup, discuss the design considerations and explain why your design is better than the previous ones
 - For theoretical work,
 - ◇ explain the theoretical framework you used
 - ◇ in deriving equations, present the most important and essential steps; you don’t need to present all the steps
 - ◇ define all the symbols you used, either within the text or as a list at the end of the report
 - ◇ in going (jumping) from one step to the next, explain the physical reasoning
 - ◇ detailed derivation or those non-essential steps should be included as an appendix to the paper
 - ◇ if you have developed a new theory, explain in greater detail the new part and why your theory is better than the previous ones.
- *Methodology, data collection and analyses*
 - For experimental work,
 - ◇ explain how you have used your equipment to collect the necessary data
 - ◇ describe the experimental considerations, including calibration procedures and possible measurement errors or problems
 - ◇ explain how you have minimized the measurement errors or addressed these problems
 - For theoretical work,
 - ◇ explain how you will use the theory to generate the necessary data
 - ◇ explain the numerical procedures you used to solve your equations.
 - ◇ discuss what you have considered in terms of the stability, convergence or other numerical problems associated with the solutions
 - For data analysis work,
 - ◇ explain the data sources and discuss possible errors in the data
 - ◇ explain why these types of data are used, and how they might be better than other datasets
 - ◇ explain and describe the approaches you have used in analysing the data
 - For each type of work,
 - ◇ describe the type(s) of analyses you have performed

- ◇ if the analysis tools are not standard, describe the tools you used and explain why they are more appropriate than the standard or conventional ones
- *Results*
 - Explain and discuss all the results you want to present. Apply your knowledge of the subject and the results of previous studies to provide physical explanations.
 - Discuss possible errors in your results, whether experimental, measurement or numerical. Include such errors (e.g. error bars) in your presentation of the results such as graphs.
 - Do not include all the data you have collected. Some of them may not be essential and can be included in an appendix. Only include the most relevant ones.
 - In presenting your results, think about what figures, tables, plates/images are necessary to illustrate and explain the results.
 - Relate them to
 - ◇ the objectives of your study
 - ◇ results of previous studies - whether they are consistent or different and explain why in the latter case; cite the appropriate references
 - ◇ your experimental design/theory and whether your theory has been verified or you have proved your design to be superior
 - Discuss any problems you have in interpreting or obtaining the results.
- *Summary/conclusion/future work*
 - The summary should be for the entire report and not just the results, i.e. from the objectives to the experimental design/theory to the methodology and analysis procedures, and then to the results.
 - Draw conclusions based on your results and discuss whether you have achieved your objectives. If not, why not.
 - Discuss problems you have encountered and how they were solved. If they could not be solved, why not and what might be done in the future.
 - Discuss any new insights you have obtained from the results.
 - Discuss any future work you plan to do.

4. Figures, plates/images and tables

GENERAL POINTS TO NOTE

- A figure is generally much better than a table in presenting a set of numbers.
- Captions should only describe the contents of the figure, plate or table. Explanations of the contents should be given in the text.
- Quote the reference if the figure, plate/image or table is adapted or copied from another publication. ***Otherwise, this is considered as plagiarism.***

FIGURES

- Label the axes of your graphs.
- Lines, curves and symbols should be explained either in the caption or with a legend.
- Avoid using dotted lines unless the dots are large.
- Use symbols that are distinct from one another and will not overwrite each other.

- Avoid too many lines or symbols on one graph.
- For contour plots, label the contours or give the contour intervals in the caption. If you want to use shading, give the colour bar of the shading. Provide units of the contour and the contour interval, and units for the shading.
- Avoid light colours as they do not show up well on a white background.
- Change the scales of the axes to maximize the display.
- Avoid presenting part of a figure on one page and the other part on the following page. The figure caption should also be on the same page.

TABLES

- Avoid too many columns and/or rows in one table
- The number of decimal places should be consistent with the measurement/analysis errors
- Put the powers of 10 in either the row/column labels or the caption.
- Avoid presenting part of a table on one page and the other part on the following page. The table caption should also be on the same page.

PLATES/IMAGES

- Make sure the image is sharp and in focus when taking the picture.
- Try to provide a scale on the plate.
- Try to use simple labelling tools (arrows, circles) to highlight the significant features.

5. Abstract

- The abstract should be a summary of the entire paper, i.e. from the objectives to the experimental design/theory to the methodology and analysis procedures, and then to the results. It is like “a summary of the summary”.
- A sentence or two on the significance of the results should also be included.

6. Appendix

- An appendix serves the purpose of providing extra information for the reader.
- Such information must not be essential for the understanding of the paper.
- Examples:
 - detailed derivation of equations
 - detailed operation of an instrument
 - description of a numerical or analysis procedure
 - additional results to supplement those in the main text

7. Spell and grammar check

Before you submit your report, you **must** perform a spell and grammar check for the entire report. While it is understood that many technical words might not be recognised by the spell check and hence are highlighted to be wrong, you should go through each of the highlighted terms and see if you indeed make a spelling mistake. If the report contains many spelling or grammatical mistakes, it implies that you are careless and you will lose marks.